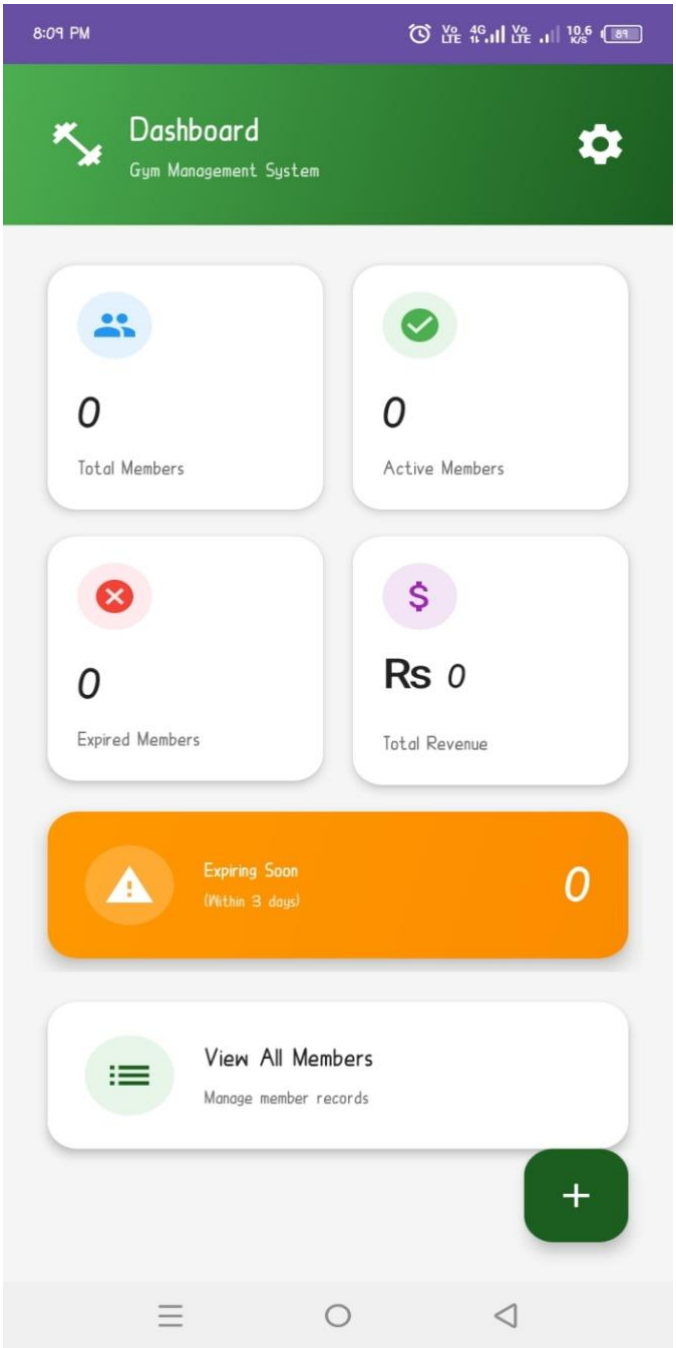
🏋️ GYM MANAGEMENT APP

Project Documentation

Platform: Android (Java + XML)

Database: SQLite

IDE: Android Studio



1️⃣ Project Overview

The Gym Management App is a fully functional Android mobile application developed for gym owners to manage members efficiently.

This application allows the gym owner (Admin) to:

Manage gym members

Track membership validity

Monitor payments and revenue

Secure the application with a PIN lock

Receive alerts for expiring memberships

The app is designed as a basic commercial-ready product and can be deployed in small to medium gyms.

2️⃣ Project Objective

To develop a secure, offline gym management system that:

Eliminates manual record keeping

Tracks active and expired members automatically

Calculates revenue dynamically

Secures access using a 4-digit PIN

Works completely offline using SQLite database

3️⃣ User Role

👤 **Admin** (Gym Owner)

Only one user role exists.

Admin can:

Set/Change App PIN

Add new members

Edit member details

Delete members

View dashboard statistics

Search members

Filter active/expired members

Receive expiry alerts

No multi-user system in this version.

4️⃣ Application Flow

🔐 Security System

If PIN is NOT set → App opens Dashboard directly

If PIN is set → App opens PIN screen first

Correct PIN → Opens Dashboard

Wrong PIN → Shake animation triggered

PIN is stored securely using SharedPreferences.

5️⃣ Core Features

5.1 Dashboard

Displays:

Total Members

Active Members

Expired Members

Total Revenue (Paid members only)

Expiring Soon members (within 3 days)

All data updates dynamically from SQLite database.

5.2 Add Member

Admin can add a member with:

- Full Name
- Phone Number
- Join Date

- Membership Plan:

1 Month

3 Months

6 Months

- Fee Amount
- Payment Status (Paid / Unpaid)
- Expiry Date is automatically calculated based on selected plan.

5.3 Member List

Displays:

- Name
- Phone
- Expiry Date
- Status (Active / Expired)

Features:

- Expired members highlighted in red
- Search by name
- Filter:

Active

Expired

5.4 Edit Member

Admin can:

Update member details

Change payment status

Modify membership plan

Expiry date recalculates automatically.

5.5 Delete Member

Confirmation dialog before deletion

Permanent removal from SQLite database

5.6 Membership Status Logic

If:

Current Date > Expiry Date → Expired

Else → Active

Status updates dynamically every time app opens.

5.7 Expiry Alerts

Members expiring within 3 days:

Highlighted on dashboard

Marked visually in member list

5.8 PIN Lock System

4-digit PIN setup in Settings

Stored using SharedPreferences

Shake animation on wrong PIN

Gradient circular number buttons

6 Bonus Features Implemented

Modern Grid Dashboard Layout

Gradient Circular Buttons

Shake Animation

Revenue auto-calculation

Clean folder structure

Input validation

Error messages

Confirmation dialogs

7 Technical Implementation

7.1 Technologies Used

- Java
- XML
- SQLite
- SharedPreferences
- RecyclerView
- Material UI Components

7.2 Database Structure

Table: members

Column Name	Type
id	INTEGER (Primary Key)
name	TEXT
phone	TEXT
join_date	TEXT
plan	TEXT
fee	REAL
payment_status	TEXT
expiry_date	TEXT

7.3 Project Structure

App/src/main

- Java

```
│  activities
│    └── PinActivity.java
│    └── DashboardActivity.java
│    └── AddMemberActivity.java
│    └── MemberListActivity.java
│    └── EditMemberActivity.java
│    └── SettingsActivity.java
│    └──SplashActivity.java
├── adapters
│    └── MemberAdapter.java
```

```
│
├── database
│   └── DatabaseHelper.java
│
├── models
│   └── Member.java
```

- res/layout

```
└── activity_dashboard.xml
└── activity_add_member.xml
└── activity_member_list.xml
└── activity_edit_member.xml
└── activity_settings.xml
└── activity_pin.xml
└── item_member.xml
└── activity_splash.xml
└── dialog_change_pin.xml
```

Each activity includes:

- Java file
- Corresponding XML layout file

Fully compatible with Android Studio.

8 UI Design

Clean professional layout

Grid-based dashboard

Gradient buttons

Red highlight for expired members

Proper spacing and alignment

Loading indicators

Error handling messages

No unnecessary animations.

9 How to Run the Project

Open Android Studio

Click Open Project

Select project folder

Let Gradle sync

Run on Emulator or Physical Device

Minimum SDK: 21+

10 Deliverables

Complete Android Project

Working APK

GitHub Repository

This Documentation

SQLite Database integrated

Fully offline working app

11 Evaluation Criteria

Functionality completeness

Code structure

SQLite implementation

UI professionalism

PIN security implementation

Bug-free execution

1 2 Future Improvements

Cloud backup option

Attendance tracking

Export to PDF

Dark mode

Member profile photo upload

Multi-admin support

Conclusion:

The Gym Management App is a secure and fully offline Android application designed to help gym owners efficiently manage members, memberships, and payments. It automates expiry tracking, revenue calculation, and member status updates using a structured SQLite database. With a modern UI and PIN-based security system, the app is ready to serve as a basic commercial gym management solution.

The End